

Lab Assignment 18.3

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Task1:

Code and prompt:

```
week18 > task1.py > get_fare
1  """
2  Write a Python script using requests to fetch public transport fare between two stations.
3  Validate user input, handle invalid station names, API downtime, and timeouts.
4  Display results in a structured format.
5  Note: Use a mock API or a public transport API that provides fare information.
6  """
7  import requests
8  def get_fare(source, destination):
9      if not source or not destination:
10         print("Invalid input: Source/Destination cannot be empty")
11         return
12         url = f"https://api.example.com/fare?source={source}&destination={destination}"
13
14     try:
15         response = requests.get(url, timeout=5)
16
17         if response.status_code == 200:
18             data = response.json()
19
20             if "fare" in data:
21                 print("\n--- Fare Details ---")
22                 print(f"From      : {source}")
23                 print(f"To        : {destination}")
24                 print(f"Fare      : ₹{data['fare']}")
25             else:
26                 print("Invalid station names or no data found.")
27
28         else:
29             print(f"API Error: {response.status_code}")
30
31     except requests.exceptions.Timeout:
32         print("Request timed out. Try again later.")
33     except requests.exceptions.RequestException as e:
34         print("Network error:", e)
35
36 # Example
37 get_fare("Hyderabad", "Warangal")
```

Task2:

Code and prompt:

```
week18 > task2.py > ...
1  """
2  Write a Python script using requests to convert INR into USD, EUR, and GBP.
3  Validate amount input and handle API errors, timeouts, and invalid responses.
4  Display output in table format.
5  """
6  import requests
7
8  def convert_currency(amount):
9      if amount <= 0:
10         print("Invalid amount")
11         return
12
13         url = "https://api.exchangerate-api.com/v4/latest/INR"
14
15         try:
16             res = requests.get(url, timeout=5)
17
18             if res.status_code == 200:
19                 data = res.json()
20                 rates = data["rates"]
21
22                 print("\n--- Currency Conversion ---")
23                 print(f"{'Currency':<10} {'Value':<10}")
24                 print("-" * 20)
25                 print(f"USD          {amount * rates['USD']:.2f}")
26                 print(f"EUR          {amount * rates['EUR']:.2f}")
27                 print(f"GBP          {amount * rates['GBP']:.2f}")
28
29             else:
30                 print("API Error:", res.status_code)
31
32         except requests.exceptions.Timeout:
33             print("Timeout error")
34         except Exception as e:
35             print("Error:", e)
36
37     # Example
38     convert_currency(1000)
```

Task3:

Code and prompt:

```
week18 > task3.py > get_repo_info
1  """
2  Write a Python function using requests to fetch GitHub repository details.
3  Handle 404 errors, rate limits, and invalid inputs.
4  Display name, description, stars, forks, and open issues.
5  """
6
7  import requests
8  def get_repo_info(owner, repo):
9      url = f"https://api.github.com/repos/{owner}/{repo}"
10     try:
11         res = requests.get(url, timeout=5)
12         if res.status_code == 200:
13             data = res.json()
14
15             print("\n--- Repository Info ---")
16             print("Name      :", data["name"])
17             print("Description:", data["description"])
18             print("Stars       :", data["stargazers_count"])
19             print("Forks       :", data["forks_count"])
20             print("Open Issues:", data["open_issues_count"])
21
22         elif res.status_code == 404:
23             print("Repository not found")
24
25         elif res.status_code == 403:
26             print("Rate limit exceeded. Try later.")
27
28         else:
29             print("Error:", res.status_code)
30
31     except requests.exceptions.Timeout:
32         print("Request timeout")
33     except Exception as e:
34         print("Error:", e)
35
36 # Example
37 get_repo_info("torvalds", "linux")
```

Task4:

Code and prompt:

```
week18 > task4.py > get_news
1  """
2  Write a Python script using News API to fetch top 5 headlines for a given category.
3  Handle invalid category, missing API key, and request failures.
4  Implement retry mechanism.
5  """
6  import requests
7  import time
8  API_KEY = "YOUR_API_KEY"
9  def get_news(category):
10     url = f"https://newsapi.org/v2/top-headlines?category={category}&apiKey={API_KEY}"
11     for attempt in range(3):
12         try:
13             res = requests.get(url, timeout=5)
14             if res.status_code == 200:
15                 data = res.json()
16                 articles = data.get("articles", [])[:5]
17                 if not articles:
18                     print("Invalid category or no news found")
19                     return
20                 print("\nTop Headlines:\n")
21                 for i, article in enumerate(articles, 1):
22                     print(f"{i}. {article['title']}")
23                 return
24             elif res.status_code == 401:
25                 print("Invalid API key")
26                 return
27             else:
28                 print("Error:", res.status_code)
29
30         except requests.exceptions.RequestException:
31             print(f"Retrying... ({attempt+1})")
32             time.sleep(2)
33
34     print("Failed after retries.")
35
36 # Example
37 get_news("technology")
```

Task5:

Code and prompt:

```
week18 > task5.py > get_covid_stats
1  """
2  Write a Python script using requests to fetch COVID-19 statistics for a country.
3  Handle invalid country names, rate limits, and network errors.
4  Include retry logic.
5  """
6  import requests
7  import time
8  def get_covid_stats(country):
9      url = f"https://disease.sh/v3/covid-19/countries/{country}"
10     for attempt in range(3):
11         try:
12             res = requests.get(url, timeout=5)
13             if res.status_code == 200:
14                 data = res.json()
15                 print("\n--- COVID-19 Stats ---")
16                 print("Country      :", data["country"])
17                 print("Total Cases  :", data["cases"])
18                 print("Deaths       :", data["deaths"])
19                 print("Recovered    :", data["recovered"])
20                 print("Active       :", data["active"])
21                 return
22             elif res.status_code == 404:
23                 print("Invalid country name")
24                 return
25             elif res.status_code == 429:
26                 print("Rate limit hit. Retrying...")
27                 time.sleep(2)
28             else:
29                 print("Error:", res.status_code)
30         except requests.exceptions.Timeout:
31             print("Timeout, retrying...")
32         except Exception as e:
33             print("Error:", e)
34
35     print("Failed after retries")
36
37 # Example
38 get_covid_stats("India")
```